SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-003: Medicine Price Validation and Role-Based Discount Management

Field	Details
Change Request ID	CR-003
System	Pharmacy Management System
Change Type	Backend and Frontend Enhancement
Priority	Not specified in source
Student Name	C. Lavasujan
Student ID	Completed
Implementation Status	Implemented (as reported)
Report Date	30 October 2026

1. Change Request Summary

1.1 Business Requirement

The Sales & Billing module requires controlled discount options that are managed by Admin users. Medicine prices and discount selections must be validated by the server so that sale totals are calculated using trusted database values rather than client-submitted prices or totals.

1.2 Existing Behaviour

The supplied CR-003 notes do not describe the previous discount-management workflow in detail. They identify the need for an approved discount list, role-based management permissions, and server-side validation of medicine prices and discounts during sale creation.

1.3 Requested Change

Provide an Admin-only Discount Management page to add and activate or deactivate approved discount options. Allow Admin, Pharmacist and Staff users to view and select active discounts during sales. Calculate item discounts and sale totals on the server using medicine selling prices retrieved from the database, reject unapproved or inactive discounts, and save price and discount snapshots for each sale item.

2. Implementation

2.1 Implementation Summary

The supplied implementation notes state that CR-003 has been implemented and that all acceptance criteria have been verified. The change adds discount-management endpoints and a dedicated Admin page, updates the Sales interface to load approved discounts, and applies database-backed price lookup, discount whitelist validation and server-side total calculation when a sale is created. The notes describe the modified files and database state, but the modified source files and execution evidence were not attached for independent inspection.

2.2 Affected Files

File	Change Type	Description
routes/sales.js	Modified	Added GET /api/sales/discounts; updated POST /api/sales for discount whitelist validation, DB price lookup and server-side calculation.
routes/admin/discounts.js	New	Admin-only GET, POST and PUT /:id/status endpoints.
app.js	Modified	Registered adminDiscountRoutes at /api/admin/discounts.
public/pages/sales.html	Modified	Added discount breakdown summary and Discounts sidebar link.
public/js/sales.js	Modified	Loads API discount options; uses dropdown; shows live item and sale breakdown; sends discount_percentage per item.
public/css/sales.css	Modified	Styles discount dropdown, item breakdown and sale summary rows.
public/pages/admin/discounts.html	New	Admin-only Discount Management page.
public/js/admin/discounts.js	New	Loads options, adds options and toggles active/inactive state.
public/css/discounts.css	New	Styles the Admin Discount Management page.
public/js/roleMenu.js	Modified	Shows the Discounts sidebar link to Admin users only.
public/pages/*.html (x7)	Modified	Added Discounts sidebar link, hidden by default and shown for Admin by roleMenu.js.
schema.sql	Updated	Documents discount_options and sale_items snapshot columns.
seed.sql	Updated	Includes five baseline discount options.

2.3 Database State

The source notes state that the schema was already applied to the current database and no migration was required.

discount_options - pre-seeded values:

ID	Percentage	Label	Status
1	0.00%	Standard (0%)	Active
2	5.00%	Special (5%)	Active
3	10.00%	Staff Discount (10%)	Active
4	15.00%	Privilege (15%)	Active
5	20.00%	VIP (20%)	Active

sale_items snapshot columns:

Column	Type	Default
discount_percentage	DECIMAL(5,2)	0.00
discount_amount	DECIMAL(10,2)	0.00

2.4 API Endpoints

Method	Route	Roles	Description
GET	/api/sales/discounts	Admin, Pharmacist, Staff	Returns active options for the Sales dropdown.
GET	/api/admin/discounts	Admin only	Returns all options, including inactive ones.
POST	/api/admin/discounts	Admin only	Adds an approved discount option.
PUT	/api/admin/discounts/:id/status	Admin only	Activates or deactivates an option.
POST	/api/sales	Admin, Pharmacist, Staff	Creates a sale with server-side validation.

2.5 Calculation Rules (Server-Side Only)

For each item:

- Gross Total = database selling_price × quantity.
- Discount Amount = round(Gross Total × discount percentage / 100, 2).
- Subtotal = Gross Total − Discount Amount.

For the sale: Final Payable = sum of all item subtotals. Client-submitted prices and totals are ignored by the server.

2.6 Implementation Logic

1. Admin manages approved discount options through the Admin Discount Management page.
2. The Sales page loads active discount options from GET /api/sales/discounts and presents them in a dropdown, with 0% as the default.
3. When a sale is submitted, the server retrieves each medicine selling price from the database.
4. The server validates each submitted discount percentage against active discount_options records.
5. The server calculates gross total, discount amount, subtotal and final payable amount independently.
6. The server stores selling_price, discount_percentage, discount_amount and subtotal as sale-item snapshot values.

2.7 Security Controls and Data Handling

Control	Implementation
Admin-only management	authorizeRoles("admin") is applied to all /api/admin/discounts routes.
Non-admin access	Management routes are protected by authorization middleware and return 403 for non-admin users.
Discount whitelist	Each submitted percentage is checked against discount_options where is_active = 1.
Inactive or unknown discounts	Inactive options are excluded; an unknown percentage is rejected with 400 Bad Request.
Database price authority	The selling_price in the medicines table is used, not the client-submitted price.
Snapshot per sale item	selling_price, discount_percentage, discount_amount and subtotal are stored in sale_items.
Frontend guard	admin/discounts.js redirects non-admin users to the dashboard.

3. Testing & Verification

3.1 Testing Approach

The supplied CR-003 notes state that all eight acceptance criteria have been implemented and verified. The test cases below translate those criteria into reportable checks. Individual test logs, test data, runtime output and screenshots were not included in the supplied source; statuses therefore reflect the source report rather than independent execution in this review.

3.2 Test Cases

ID	Test Case	Expected Result	Actual / Status
TC-01	Admin adds or manages a discount option.	Only Admin can manage discount options.	Reported verified
TC-02	Pharmacist or Staff opens Sales and selects a discount.	Can view/select options but cannot manage them.	Reported verified
TC-03	Open Sales discount dropdown.	Only active options appear; default is 0%.	Reported verified
TC-04	Change the discount dropdown selection.	Item discount and subtotal update live.	Reported verified
TC-05	Submit an unapproved, inactive, duplicate or out-of-range discount.	Invalid discount selection is rejected.	Reported verified
TC-06	Submit a sale with client price or total values.	Server uses database prices and ignores client prices/totals.	Reported verified
TC-07	Create a sale with a discount and inspect sale item data.	Required price and discount snapshot fields are retained.	Reported verified
TC-08	Run the existing sales, authentication and stock workflow.	Existing workflow remains unchanged.	Reported verified

Test summary: 8 acceptance-related checks are reported as verified in the source notes. The source does not provide separate pass/fail execution records.

4. Acceptance Criteria Verification

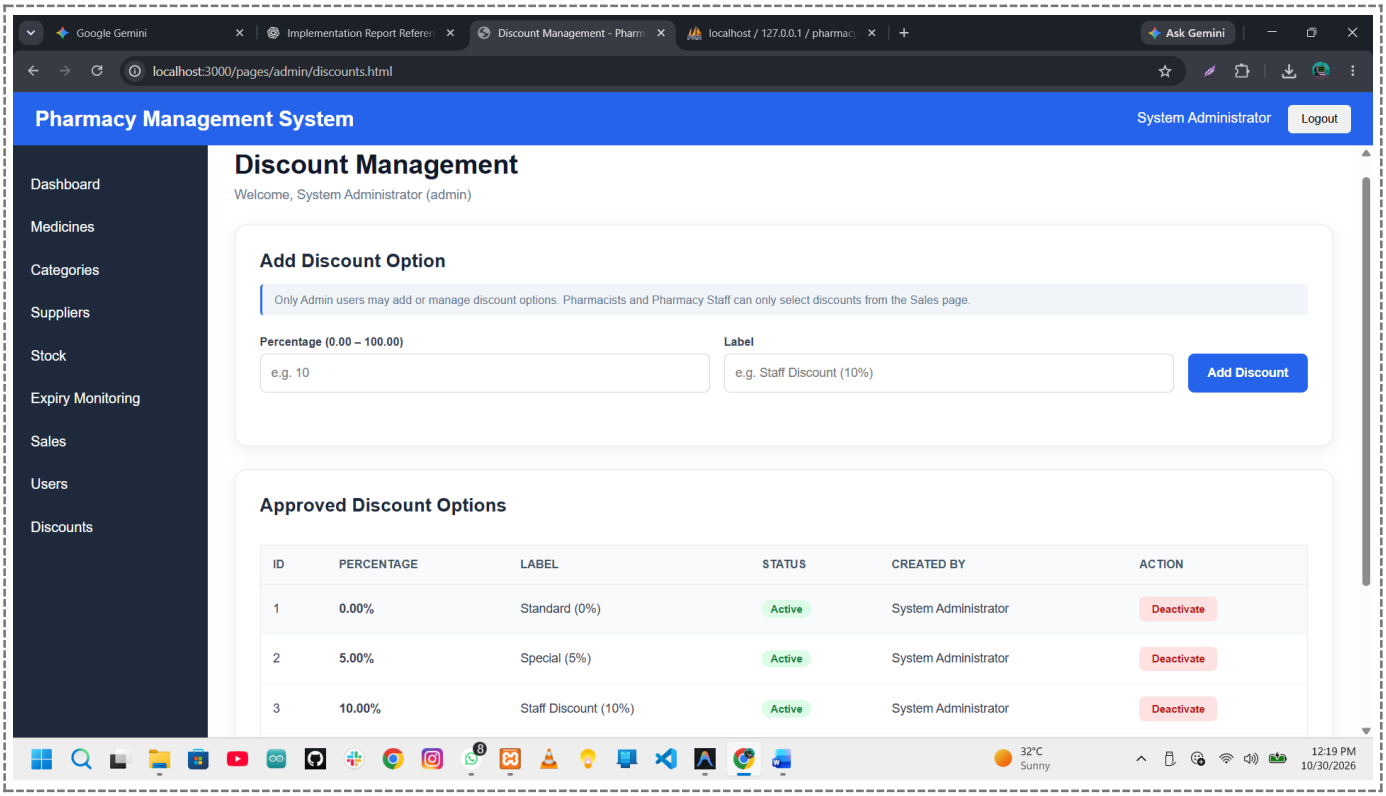
ID	Criterion	Status
AC-001	Only Admin can add/manage discount options.	Reported verified
AC-002	Pharmacist/Staff can view and select, but cannot manage options.	Reported verified
AC-003	Sales dropdown shows active options only and defaults to 0%.	Reported verified
AC-004	Changing dropdown updates discount and subtotal live.	Reported verified
AC-005	Unapproved, inactive, duplicate and out-of-range values are rejected.	Reported verified
AC-006	Backend uses database prices and ignores client-submitted prices.	Reported verified
AC-007	Sale items retain price and discount snapshot fields.	Reported verified
AC-008	Existing authentication, stock and sales workflow remains unchanged.	Reported verified

5. Evidence

The CR-003 source file does not include screenshots or other visual evidence. Insert the corresponding application screenshots in the reserved spaces below before final submission.

Evidence 01 - Admin Discount Management

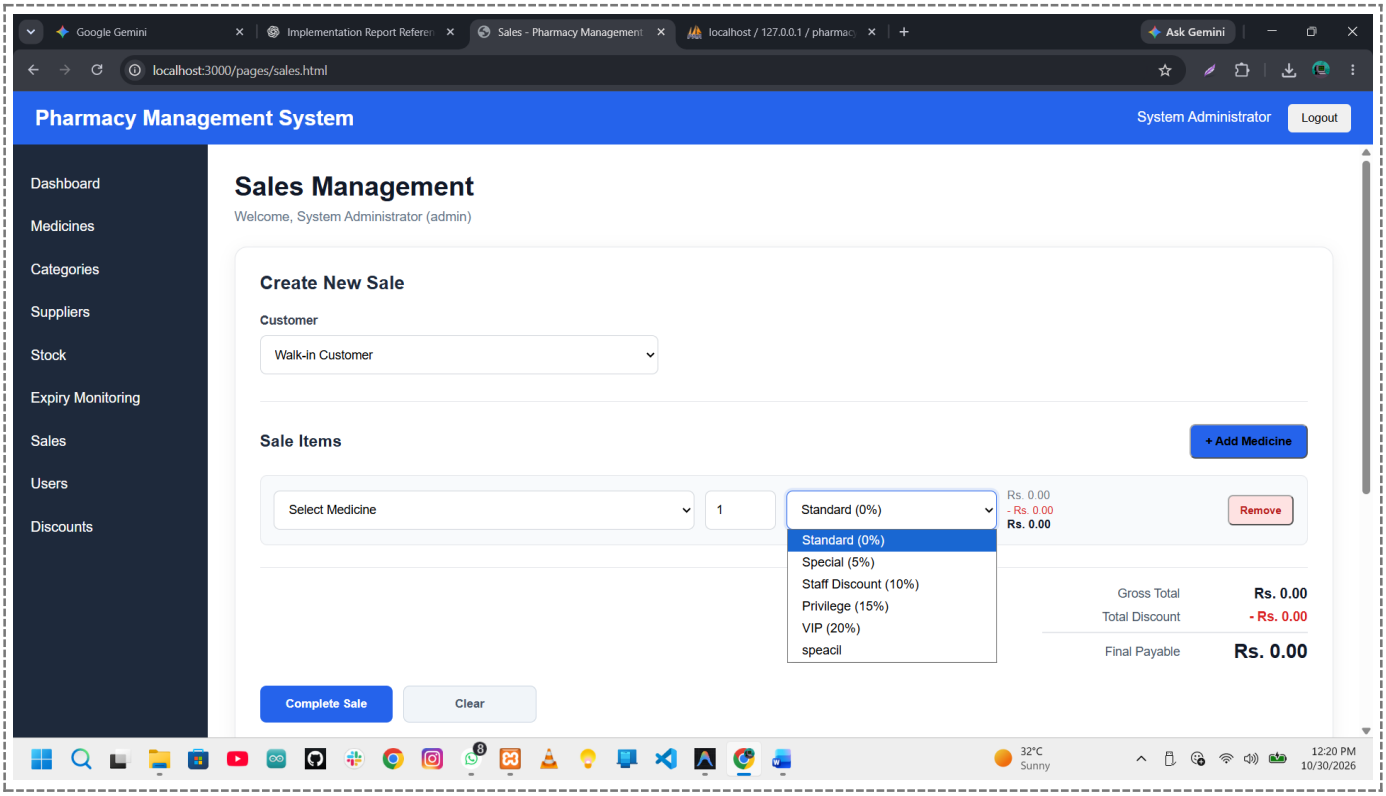
Admin Discount Management page showing the approved discount list and controls for adding or changing active status.



Related: AC-001.

Evidence 02 - Sales Discount Selection and Breakdown

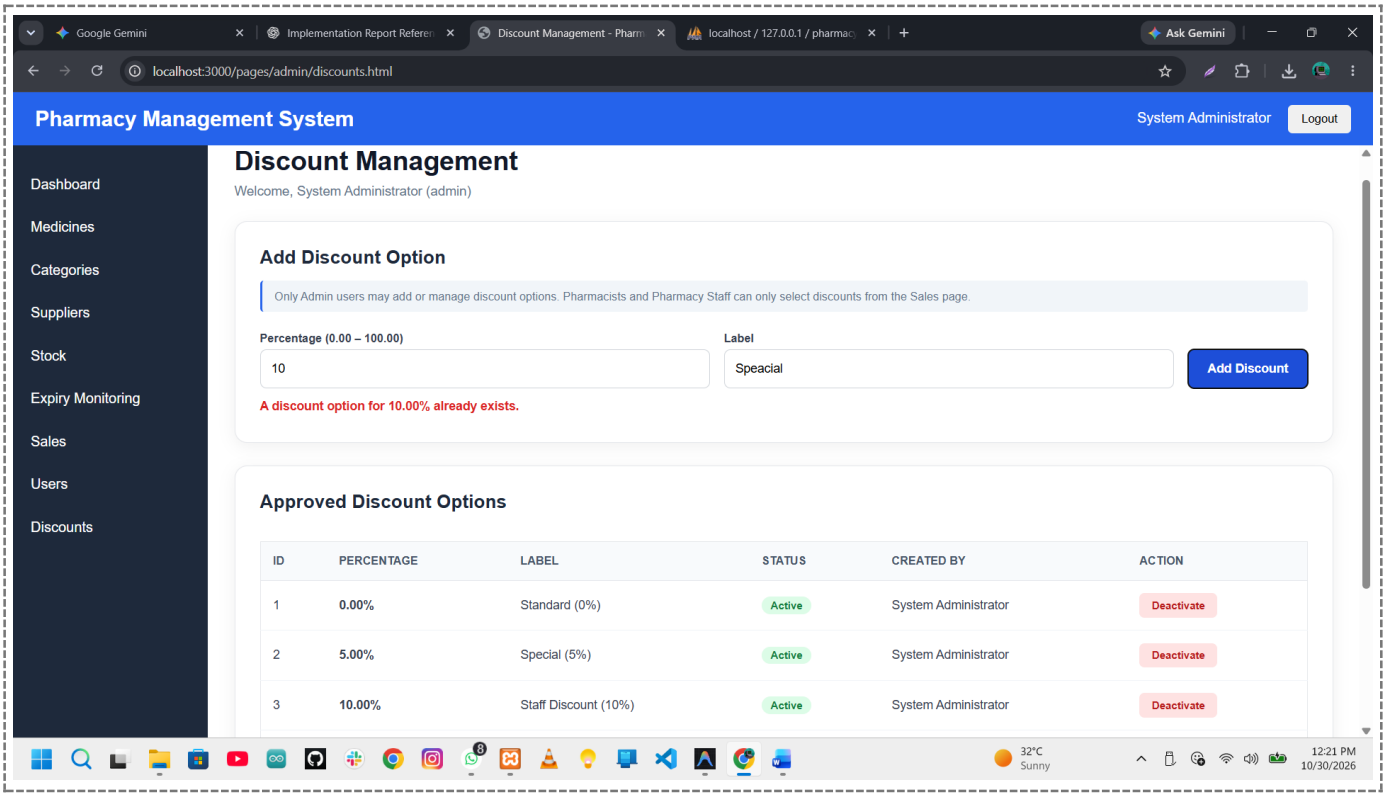
Sales page showing active discount options and the live per-item and sale-total discount breakdown.



Related: AC-002 to AC-004.

Evidence 03 - Validation and Sale Item Snapshot

Evidence of rejected invalid discount submissions and saved sale-item snapshot values.



Related: AC-005 to AC-008.

6. Final Result

Item	Result
Requirements	Implemented (as reported)
Admin Discount Management	Implemented (as reported)
Server-side Price Validation	Implemented (as reported)
Discount Whitelist Validation	Implemented (as reported)
Sale Item Snapshots	Implemented (as reported)
Acceptance Criteria	8 criteria reported verified
Final Status	Implemented (as reported)

6.1 Final Summary

CR-003 introduces Admin-controlled discount options and server-side validation for the Sales & Billing module. Active discount values are supplied through the API, role permissions restrict discount management to Admin users, and sale calculations use medicine prices retrieved from the database. The final payable amount is calculated on the server, while price and discount details are retained in sale_items for each sale line. The source report states that all eight acceptance criteria have been implemented and verified.

6.2 Issues / Limitations

The supplied report does not include the modified application source files, individual test execution records, or screenshots. Accordingly, implementation and verification statuses in this document are attributed to the supplied CR-003 notes and have not been independently reproduced. Evidence placeholders remain to be completed with screenshots from the running system.

6.3 Out of Scope

- Customer loyalty points (CR-008).
- Multi-medicine basket promotions, such as Buy 2 Get 1 Free (CR-009).
- Multi-currency support; the system uses Rs. only.

6.4 Deployment Notes

The discount_options table and the new sale_items columns must exist before deployment. The source notes state that these are already applied in the current environment. For a fresh installation, run schema.sql followed by seed.sql. The stated deployment order is: Database -> Backend -> Frontend -> regression test all three roles.

Student Name: _____

Student ID: _____

Date: _____

Signature: _____